skaters: Model First, Conform Last — Composable Online Distributional Time-Series Forecasting in Python and JavaScript

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Abstract. We present **skaters**, a zero-dependency library for online univariate time-series forecasting in which *every prediction is a full probability distribution*, not a point. Models are built by composition: invertible transforms chain together, ensembles nest, and a single distributional *leaf* sits at the bottom of every chain. The leaf is a pluggable proper-scoring-rule optimiser, which lets us separate two concerns that the forecasting and conformal-prediction literatures usually conflate — *modelling* the conditional mean (judged by likelihood) and *calibrating* the predictive tail (optionally judged by a downstream score such as the continuous ranked probability score, CRPS). We call the resulting recipe **model first, conform last**, and show that it matches a dedicated CRPS specialist on CRPS while *improving* held-out log-likelihood. In a fair rolling one-step-ahead comparison on 500 FRED change-series against *eight* distributional baselines — **AutoARIMA**, **AutoETS**, **statsmodels SARIMAX** and exponential smoothing, a **GARCH**$(1,1)$ model with Student-t innovations, a **NeuralForecast** model with a Student-t head, and **AutoARIMA** paired with split-conformal and adaptive-conformal residuals — the general forecaster **wins the per-series log-likelihood race against all eight**. The decisive test is the **GARCH**-t model, the classical heavy-tail / volatility-clustering state of the art: our forecaster still edges it on the continuous subset (mean continuous log-likelihood 2.86 versus 2.72), confirming the heavy-tail advantage *against the heavy-tail specialist.* On CRPS the picture is mixed, as expected: we beat the smoothing, **GARCH** and neural baselines, roughly tie the **ARIMA** family, and lose to the CRPS-optimised conformal variants. We further contribute a *mean-preserving lattice projection* for series that revisit exact values, an online *coordinate* (Yeo–Johnson) search, and a committed *martingale* model whose volatility clock is a time-changed Brownian motion. The whole library is implemented twice — pure Python and zero-dependency JavaScript — and verified identical to 10^{-6} on roughly 90,000 probe values, so the same models run server-side or in the browser via **Pyodide**.

Keywords: probabilistic forecasting, time series, proper scoring rules, conformal prediction, online learning, Python, JavaScript, **skaters**.

1 Introduction

Most deployed forecasters return a point and, perhaps, an interval bolted on afterwards. Yet the quantity a decision-maker actually needs is the *predictive distribution*: the mean for an estimate, the spread for risk, the tails for position sizing, and a density for likelihood-based evaluation and combination. Two strands of recent work take distributions seriously but each gives something up.

Classical state-space and exponential-smoothing models [Hyndman et al., 2002, Hyndman and Khandakar, 2008] are distributional but rarely *online and composable*: adding a transform usually means re-deriving the filter. Conditional-heteroscedasticity models [Engle, 1982, Bollerslev, 1986] capture the heavy tails and volatility clustering of financial data, but commit to a single parametric form and are refit in batch. At the other extreme, **conformal prediction** [Vovk et al., 2005, 2019] is distribution-free and online, but a conformal predictive system outputs a *cumulative distribution function*, not a density. It is calibrated for coverage and CRPS but **cannot be scored on log-likelihood at all**, and an empirical conformal CDF assigns zero density — and hence $-\infty$ log-likelihood — to any value outside the observed residual range. Modern neural and foundation models [Salinas et al., 2020, Ansari et al., 2024, Das et al., 2024, Woo et al., 2024] are powerful distributional learners, but are built for global cross-series training on hardware accelerators, not for a single univariate stream running in a browser.

skaters takes a third route. Every node — transform, ensemble, or leaf — is a function `(y, state) -> ([Dist], state)` that consumes one observation and emits a list of `Dist` objects, one per forecast horizon, where a `Dist` is a weighted Gaussian mixture. Because the type is closed under composition, an entire model is a chain of invertible transforms with one distributional leaf at the bottom, and ensembles are simply skaters that combine other skaters. This single abstraction yields three contributions developed in turn:

1. **A pluggable objective** (Sections 3–4). The leaf fits its mixture weights by *a* proper scoring rule. Point it at log-likelihood and it models honestly; point its *terminal* copy at CRPS while keeping a likelihood-weighted trunk, and you get **model first, conform last** — competitive with a CRPS specialist on CRPS *and* better on likelihood.

2. **Distributional structure that survives composition** (Sections 5–7). A *mean-preserving lattice projection* places atoms on the exact values a series revisits without moving the ensemble mean and while vanishing on continuous data; an online *coordinate* search learns whether a series is simple in a log, root, or linear coordinate; and a committed *martingale* model learns only a volatility clock.

3. **An honest benchmark against eight distributional baselines** (Section 8), including the heavy-tail (**GARCH-*t***) and neural (**NeuralForecast**) state of the art, scored on held-out log-likelihood and CRPS in a single, identical `Dist`-based protocol.

The unifying premise is the No-Free-Lunch theorem [Wolpert and Macready, 1997] taken seriously: every "method" is a prior, and averaged over all series none dominates. The right response is not to pick one but to make the priors explicit, weight them online, and bound the cost of a wrong prior by how fast the weighting abandons it.

2 The `Dist` abstraction and composable transforms

A `Dist` is a weighted mixture $\sum_i w_i \mathcal{N}(\mu_i, \sigma_i^2)$ with $\sigma_i > 0$ and $\sum_i w_i = 1$. It exposes `mean`, `std`, `pdf`, `cdf`, `logpdf`, `quantile`, and a closed-form `crps`, together with the affine maps (`shift`, `scale`, `affine`) and a moment-matching `prune` needed to propagate it through transform inverses and to bound component growth in ensembles. The Gaussian mixture is a deliberate choice: it is closed under affine maps and mixing, admits a closed-form CRPS through the elementary identity $\mathbb{E}\,|\mathcal{N}(m, s^2)| = 2s\,\phi(m/s) + m\,(2\Phi(m/s) - 1)$, and approximates heavy tails arbitrarily well as a scale mixture.

A *transform* is an online bijection given by a `forward` (scalar in, scalar out) and an `inverse_k` that maps k `Dist`s in the transformed space back to the original. Differencing, exponential smoothing, drift, Holt's linear trend, fractional differencing, autoregression (via online recursive least squares), **GARCH**-style scaling, seasonal differencing, power and Yeo–Johnson coordinates [Box and Cox, 1964, Yeo and Johnson, 2000], and standardisation are all transforms. *Conjugation* composes a transform with an inner skater,

$$y \xrightarrow{T} y' \to \text{inner} \to D' \xrightarrow{T^{-1}} D,$$

and because every step is online and the inverse propagates the full mixture, multi-step uncertainty accumulates correctly — for example variance grows as $\sum_h \sigma_h^2$ under differencing.

Numerical note. Mixture variance is computed in centred form, $\mathrm{Var} = \sum_i w_i \, (\sigma_i^2 + (\mu_i - \bar{\mu})^2)$, rather than $\mathbb{E}[X^2] - \mathbb{E}[X]^2$. The latter catastrophically cancels when a tight component sits at a large mean, silently producing $\sigma = 0$ and an invalid predictive. Running moments use Welford's recurrence [Welford, 1962].

3 The terminal-leaf ensemble: model first

Bayesian model averaging over a heterogeneous pool [Hoeting et al., 1999, Raftery et al., 2005] combines full predictive distributions, which preserves the combined mean and variance but *washes out higher moments*: a heavy-tailed leaf used inside the pool collapses toward a Gaussian shape at the output, because a mixture of differently located Gaussians is dominated by the spread of their means. We therefore use the candidate models only as **mean forecasters**, weight them online by predictive likelihood (with a learning-rate shrinkage and a per-depth complexity penalty, in the spirit of gradient boosting), combine their means, and model the distribution of the *combined residual* with a **single terminal leaf**:

$$y \;\to\; \hat{\mu} = \sum_i w_i \, \hat{\mu}_i, \qquad r = y - \hat{\mu} \;\to\; \text{leaf} \;\to\; D, \qquad \hat{D} = D \text{ shifted by } \hat{\mu}.$$

Because exactly one leaf reaches the output, its shape — heavy tails, learned online — survives undiluted. This is the *model first* half of the recipe: a likelihood-weighted trunk whose only job is to get the conditional mean right. The candidate pool spans exponential smoothing, differencing, drift, Holt's linear trend, autoregression, grouped long-lag autoregression, fractional differencing, a seasonal block, a Yeo–Johnson coordinate grid (Section 6), and a fast/slow two-systems block.

4 Metric-agnostic leaves: conform last

The leaf is itself a fixed Gaussian *scale mixture* — components $\mathcal{N}(0, c_k \, \hat{\sigma})$ over a fixed log-spaced dictionary $\{c_k\}$, with $\hat{\sigma}$ an exponentially weighted scale — whose only learned quantities are the mixture weights $\{w_k\}$ on the simplex. A Gaussian scale mixture is a Student-t in the limit, so the leaf approximates heavy tails by construction while remaining a plain `Dist`. The weights are fit online by **a proper scoring rule** [Gneiting and Raftery, 2007]:

- `likelihood` — recency-weighted expectation–maximisation on the component responsibilities;

- `crps` — exponentiated-gradient descent on the simplex minimising the closed-form mixture CRPS [Matheson and Winkler, 1976]; the gradient is exact, since the mixture CRPS is a finite sum of $\mathbb{E}\,|\mathcal{N}(m, s^2)|$ terms.

Placing the CRPS leaf at the *terminal* of a likelihood-weighted trunk gives **model first, conform last**. The trunk's job is to model, and the honest objective for modelling is likelihood — a proper, tail-sensitive score that is, up to a constant, the Kelly log-growth criterion [Kelly, 1956]; corrupting it to chase CRPS would repeat the conformal mistake of discarding the density. The tail's job is to be shaped, and that is the only place a downstream metric (CRPS) should get a vote. Section 9 shows this configuration dominates a pure-likelihood policy on CRPS *and* on likelihood: a CRPS-fit leaf is more outlier-robust, so it even generalises slightly better in log-likelihood. The objective is a knob, not a fixed cost.

5 The lattice projection

Administrative and grid-quoted series — policy rates, posted prices, pegged exchange rates — revisit a small set of exact values, and a continuous predictive density cannot place mass there. We add a *projection*: a recency-weighted frequency table of exact values, with near-Dirac atoms on the values revisited above a noise floor, each carrying its observed frequency as probability. The atom is **mean-preserving**: the continuous part is recentred so the atoms add *mass* without moving the ensemble mean.

Two properties make this a safe default. First, on continuous data nothing is revisited, no atom fires, and the projection vanishes identically — it is free when unused. Second, unlike a last-value spike it captures values that recur often but never consecutively (a rate that repeatedly returns to a round number). The pull of the level (its tendency to persist) is left to the trunk, where the random-walk candidate earns its weight by likelihood, so the projection remains a pure statement about mass, not mean.

6 Coordinate learning

Whether a series is simple in a linear, log/multiplicative, or root coordinate is itself learnable. We add a coarse grid of **Yeo–Johnson** λ candidates — the signed Box–Cox family [Box and Cox, 1964, Yeo and Johnson, 2000], defined on all of $\mathbb{R}$ — to the pool, so the ensemble learns the coordinate online rather than committing up front. This is No-Free-Lunch-safe: a wrong coordinate is simply down-weighted by likelihood. The same mechanism expresses a non-negativity prior at $\lambda = 0$ (the log coordinate), though we note that because the `Dist` is a Gaussian mixture the inverse uses the delta-method linearisation, so non-negativity is approximate rather than exact.

7 A martingale with a learned volatility clock

For near-martingale *levels* we provide a committed model, `doob`: the mean is pinned to the last value (a driftless martingale) and only the volatility clock is learned, as a Bayesian average over martingale predictives that differ in their volatility model (constant, **GARCH**, slowly varying, heavy-tailed). Because every candidate shares the same mean, plain averaging blends the clocks *without* washing out kurtosis — the committed mean is exactly what makes model averaging the right tool here, in contrast to Section 3. By the Dambis–Dubins–Schwarz theorem [Dambis, 1965, Dubins and Schwarz, 1965, Doob, 1953] a continuous martingale is a time-changed Brownian motion, so the model is literally

“Brownian motion on a stochastic clock”. It beats the general forecaster on near-martingale levels by committing the mean and spending its capacity on the clock; on genuinely mean-reverting series the martingale prior is wrong and it gives ground — a deliberately sharp instrument, fed *levels* rather than pre-differenced changes.

8 Experiments

Universe. To avoid a hand-picked series list we draw from the Federal Reserve Economic Data [Federal Reserve Bank of St. Louis], taking cached daily series and auto-transforming each to its one-step *changes* (log-difference for strictly positive levels, else first difference) — the stationary-ish innovation stream where heavy tails and regime shifts live, and which keeps log-likelihood comparable across series of different scale. We keep the last 1000 changes per series and score the last 300 as a rolling one-step-ahead test window after a burn-in.

8.1 Against eight distributional baselines

On 500 such series we compare the general forecaster `laplace` against every distributional one-step baseline we could assemble: **statsforecast**'s [Garza et al., 2022] **AutoARIMA** and **AutoETS**; **statsmodels SARIMAX**$(1, 0, 1)$ and simple exponential smoothing, which emit *exact* closed-form Gaussian predictives; a constant-mean **GARCH**$(1, 1)$ model with Student-t innovations [Bollerslev, 1986, 1987] from the **arch** package, the classical heavy-tail / volatility-clustering state of the art; a **NeuralForecast** [Olivares et al., 2022] multilayer perceptron with a Student-t distribution head; and **AutoARIMA** paired with conformal residual quantiles [Vovk et al., 2019] in both a split-conformal and an adaptive-conformal (ACI, Gibbs and Candès, 2021) variant.

The protocol is a *fair rolling one-step-ahead*: each baseline is fit on an expanding or rolling window with periodic refit, and every method — ours and theirs — is turned into the *same* `Dist` and scored on held-out log-likelihood *and* CRPS over the same test window. The parametric Student-t predictives (**GARCH**, **NeuralForecast**) are represented as finite Gaussian scale mixtures, matching the analytic t density to $\sim 10^{-3}$. We report **per-series** win-rates (the fraction of series on which `laplace` scores better), and additionally restrict to the 309 *continuous* series — those with under 5% exactly-repeating one-step changes — since on repeating/grid series the lattice projection (Section 5) gives a large but metric-specific advantage that would otherwise dominate the aggregate. Table 1 reports the result.

The likelihood race. `laplace` wins the per-series log-likelihood race against *all eight* baselines, on the full universe and on the continuous subset, including the two exact-Gaussian references (**SARIMAX**, statsmodels ETS) for which there is no band-reconstruction approximation to blame.

GARCH-t is the decisive test. The **GARCH**$(1, 1)$-t model is purpose-built for the heavy tails and volatility clustering of financial change-series, so it is *supposed* to be the hardest opponent on log-likelihood — and it is. It is the tightest race (67% of continuous series), and its mean continuous log-likelihood, 2.72, is closest to ours, 2.855 — a genuine but *narrow* edge of about 0.14 nats per observation. That the advantage survives *against the heavy-tail specialist* is the result that matters: the general forecaster captures the same conditional heteroscedasticity without committing to a parametric volatility law.

Baseline	LL (all / cont.)	CRPS (all / cont.)	Mean LL (cont.)
AutoARIMA	**78 / 64 %**	51 / 47 %	2.09
AutoETS	**92 / 88 %**	68 / 67 %	2.20
SARIMAX (statsmodels)	**82 / 72 %**	53 / 49 %	2.13
ETS (statsmodels)	**93 / 89 %**	69 / 68 %	2.12
GARCH$(1, 1)$-t	**79 / 67 %**	76 / 66 %	**2.72**
NF-Student$t$	**100 / 100 %**	78 / 76 %	0.82
AutoARIMA + conformal	**84 / 75 %**	37 / 29 %	1.47
AutoARIMA + ACI	**87 / 80 %**	36 / 28 %	1.89

Table 1: Per-series win-rate of `laplace` versus each baseline on 500 FRED change-series, all and on the 309 continuous series. LL counts series where `laplace` has the higher held-out log-likelihood; CRPS where it has the lower CRPS. The final column is each baseline's mean continuous log-likelihood (`laplace`: **2.855**). Higher is better for LL; lower for CRPS.

The neural result is not a scalp. `laplace` beats NF-Student$t$ on 100% of series, but we are deliberately loud about what this does and does not mean. NF here is **NeuralForecast** run in the *matched online univariate one-step* protocol — a tiny per-series multilayer perceptron with periodic refit — which is its **weak** regime. Neural and foundation forecasters [Salinas et al., 2020, Olivares et al., 2022] are built for *global cross-series* training, where they amortise structure across thousands of related series. Our result says nothing about **DeepAR**-style models in that home setting, and we do not claim it does; it says only that a single short univariate stream does not give a neural density head enough to beat an online ensemble. (We verified the NF densities are well formed — median log-likelihood 0.50, none floored — so the 100% is a real calibration gap, not a degenerate score.)

CRPS. As ever, CRPS tells a mixed story. `laplace` beats the smoothing baselines (**AutoETS**, ETS), **GARCH**-t and NF on CRPS, roughly ties the **ARIMA** family (**AutoARIMA** 51%, **SARIMAX** 53%), and *loses* to the conformal variants (37%, 36%), which are CRPS-optimised by construction. This is exactly the thesis of Sections 4: likelihood is the metric where a faithful density wins; CRPS is conformal's home turf, where we are competitive but not dominant. A practitioner who truly only cares about CRPS can repoint the leaf (Section 9); a practitioner who cares about the density — for risk, sizing, or combination — should prefer the honest log-likelihood column.

Scope. Five hundred change-series, one-step horizon. Prophet, multi-step horizons and raw levels are left for later; zero-shot foundation models get their own protocol in Section 8.2.

8.2 Against zero-shot foundation models

Pretrained time-series foundation models use a fundamentally different protocol: they are applied *zero-shot*, conditioning on a context window with no fitting. We evaluate four — Chronos-Bolt [Ansari et al., 2024], TimesFM 2.5 [Das et al., 2024], Moirai [Woo et al., 2024], and Lag-Llama [Rasul et al., 2023] — on 120 change-series (69 continuous). At each step the model receives a fixed 256-length window of the preceding changes and predicts the next change, with all windows for a series batched into one inference call. Each

Model	density	LL (all / cont.)	CRPS (all / cont.)	Mean LL (cont.)
Moirai (1.1-R-small)	native t	98 / **97** %	94 / 93 %	0.92
Lag-Llama	native t	67 / **99** %	65 / 94 %	0.39
Chronos-Bolt (small)	quantile*	76 / **100** %	67 / 88 %	−1.96
TimesFM (2.5-200M)	quantile*	75 / **100** %	58 / 72 %	−1.18

Table 2: Per-series win-rate of `laplace` versus four zero-shot foundation models on 120 change-series (69 continuous). *Chronos-Bolt/TimesFM log-likelihood is reconstructed from a quantile head and is tail-limited; read their CRPS as the fairer signal. `laplace` mean continuous logpdf: **1.51**.

predictive is turned into the same `Dist` — the native Student-t head for Moirai/Lag-Llama, a sample/quantile reconstruction for Chronos-Bolt/TimesFM — and `laplace` is re-scored on the identical window. Table 2 reports per-series win-rates.

On the continuous series, `laplace` beats all four foundation models zero-shot on both metrics (97–100% LL, 72–94% CRPS). The native-density models (Moirai, Lag-Llama) are the meaningful likelihood opponents and the closest, but still lose per-series. On repeat-heavy/grid series the foundation models are competitive or better — Lag-Llama even beats `laplace` on mean log-likelihood over the *full* universe (6.54 vs 4.52) by placing tight mass on revisited values, the same effect as the lattice projection — which is why the continuous split matters. The honest scope is narrow: zero-shot, no refit, fixed context, and forecasting the change stream rather than the levels these models were chiefly trained on. Whether per-series *fine-tuning* closes the gap is a separate study; for the larger models it is GPU/MPS-bound.

9 Ablations: repointing the objective

A companion sweep on 2,501 FRED series isolates the effect of the terminal-leaf objective against the **crepes** conformal predictive system (given its three best calibration windows and scored on its own CDF via the pinball decomposition of CRPS). Because the conformal opponent there uses only a naive mean, we read this as an ablation of our own objective knob rather than a state-of-the-art claim. Reporting per-policy and de-correlating to one vote per correlated family (195 families):

Forecaster (identical structure)	CRPS, raw	CRPS, family	Mean LL
likelihood trunk + likelihood leaf	82.6 %	48.7 %	2.96
likelihood trunk + CRPS leaf (*model first, conform last*)	89.5 %	**60.1 %**	**3.01**
bare CRPS leaf (no trunk)	80.8 %	60.3 %	2.96

Table 3: Effect of the terminal-leaf objective. A general likelihood policy is essentially even with conformal on CRPS over independent families (48.7%); repointing only the terminal leaf to CRPS lifts that to 60.1% *and* raises log-likelihood, matching the dedicated CRPS specialist while modelling honestly.

Three further observations motivate the radically small public API. **Priors wash out:** across 25 series with ≥ 2000 changes the conventional prior-flavoured policies (trend, long-memory, drift, minimax, fast/slow) sit within 0.0006 nats of the uniform-prior general forecaster on the second half of each series, and the first-half winner beats the general forecaster on the second half only 44% of the time — worse than chance. **Conform-last is near-free:** on idealised light-tailed processes the CRPS leaf costs 0.005–0.007 nats versus the likelihood leaf, while on heavy-tailed and stochastic-volatility processes it *helps* (up to +0.04); real data lives in the second regime. **The lattice projection is free when unused.** Together these justify exposing exactly *one* general forecaster (uniform prior, CRPS leaf, lattice projection and coordinate learning on by default) plus *one* committed martingale specialist; the remaining structural ideas do not pay their way out of sample.

10 Discussion

The unifying theme is No-Free-Lunch [Wolpert and Macready, 1997] taken seriously. Every "method" is a prior; averaged over all series none dominates. The correct response is not to choose one but to make the priors explicit, weight them online, and bound the cost of a wrong prior by how fast the weighting abandons it. Read that way, the named methods of classical forecasting are convenience bundles, and the honest object is a single adaptive ensemble with two orthogonal terminal knobs — *which proper scoring rule the leaf optimises*, and *whether a lattice projection is active* — neither of which is a prior over the trunk. The empirical headline supports the design: against the strongest distributional opponents available on CPU, including the heavy-tail and neural state of the art, the general forecaster wins the likelihood race, and where it does not lead — CRPS — the deficit is confined to methods explicitly optimised for that metric and is recoverable by repointing a single knob.

11 Heritage and reproducibility

skaters is a from-scratch rewrite distilling ideas from **timemachines** [Cotton, 2021], an earlier competition-tested online-forecasting package, and builds on years of running live distributional-prediction contests at Microprediction, where forecasts are scored continuously as full distributions. The library is implemented twice — pure Python and zero-dependency JavaScript — and a parity suite checks roughly 90,000 probe values to 10^{-6} on every run, so results are reproducible across both runtimes and in the browser via **Pyodide** [The Pyodide development team, 2021]. The benchmark of Section 8 is a single parallel script (`benchmarks/sota_study.py`) that scores every method through the identical `Dist` interface; given a FRED key and the optional baselines (**statsforecast**, **arch**, **statsmodels**, **neuralforecast**) it reproduces Table 1 end to end.

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